AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

Nominal and Effective Interest Rates

Engineering Economy

Chapter 4

What you see is NOT what you get



Example

- Consider a principal of $100 to be invested for one year at an interest rate of 1% per month.
  1. What is the annual interest rate? (quick answer)
  2. What is the amount of interest?
  3. Compute the annual interest rate?

$$Annual\ interest\ rate = 12 \times 1 = 12\%$$

$$F = P(F/P, 0.01, 12) = 100(1 + 0.01)^{12} = 112.68$$

Amount of interest is $112.68 - 100 = 12.68$

Computed Annual interest rate: $\frac{12.68}{100} = 12.68\%$



Nominal Interest Rate

- Referred to as r

The rate that does **NOT** consider compounding

- Nominal Rate ←→ Annual Percentage Rate (APR)

- Frequently reported per year



Nominal Interest Rate

- Consider a principal of $100 to be invested for one year at an interest rate of 1% per month.
  1. What is the annual interest rate? (quick answer)
  2. What is the amount of interest?
  3. Compute the annual interest rate?

Nominal Interest Rate r

$$Annual\ interest\ rate = 12 \times 1 = 12\%$$

$$F = P(F/P, 0.01, 12) = 100(1 + 0.01)^{12} = 112.68$$

Amount of interest is $112.68 - 100 = 12.68$

Computed Annual interest rate: $\frac{12.68}{100} = 12.68\%$

Effective Interest Rate

- Referred to as i

The rate that **does** consider compounding

- Effective Rate ←→ Annual Percentage Yield (APY)

Effective rate per compounding period

$$i = \frac{r}{m}$$

Nominal interest rate

Compounding frequency ($m = 1/CP$)

Effective Interest Rate

- Consider a principal of $100 to be invested for one year at an interest rate of 1% per month.
  1. What is the annual interest rate? (quick answer)
  2. What is the amount of interest?
  3. Compute the annual interest rate?

Nominal Interest Rate r

Effective Interest Rate i

$$Annual\ interest\ rate = 12 \times 1 = 12\%$$

$$F = P(F/P, 0.01, 12) = 100(1 + 0.01)^{12} = 112.6$$

Amount of interest is $112.68 - 100 = 12.68$

Computed Annual interest rate: $\frac{12.68}{100} = 12.68\%$

Directly finding i

- Let i_a be the annual effective interest rate,

$$i_a = \left(1 + \frac{r}{m}\right)^m - 1$$

Example: $r = 12\%, m = 12$, then:

$$i_a = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = 1.01^{12} - 1 = 12.68\%$$



Factors under m-time-a-year Compounding

Chapter 2

Notation

$(X/Y, i, n)$

- Find X amount, given:
  - Y amount
  - $i\%$ interest per CP
  - n number of periods
- P: Present Worth
- F: Future Worth
- A: Annual (periodic) Amount (Uniform Series)
- G: Arithmetic Gradient
- g: Geometric Gradient



5

Factors under m-time-a-year Compounding

$(X/Y, r, m, n)$

- Find X amount, given:
  - Y amount
  - $r\%$ nominal interest
  - m compounding frequency
  - n number of periods

- P: Present Worth
- F: Future Worth



Factors under m-time-a-year Compounding

$(X/Y, r, m, n)$ $\Leftrightarrow$ $(X/Y, \mathrm{i}, n)$

- Find X amount, given:
  - Y amount
  - $r\%$ nominal interest
  - m compounding frequency
  - n number of periods

- P: Present Worth
- F: Future Worth

$$(P/F, r, m, n) = \frac{1}{(1 + r/m)^n}$$



Compounding Periods and Frequencies

Name	CP	CP (Fraction of year)	Yearly Frequency (m)
Yearly	1	1	1
Semi-annually	6 months	1/2	2
Quarterly	3 months	1/4	4



Continuous Compounding

- When Compounding Period (CP) → 0
  compounding frequency (m) →∞

$$i = e^r - 1$$

- i is the effective annual interest rate
- Factors can be deduced by replacing i with $e^r - 1$



Continuous Compounding - Factors

Name	Notation	Equation
Compound Amount	$(F/P, r, t)$	e^{rt}
Present Worth	$(P/F, r, t)$	e^{-rt}
Uniform Series Present Worth	$(P/A, r, t)$	$\frac{e^{rt}-1}{e^{rt}(e^{r}-1)}$
Uniform Series Compound Amount	$(F/A, r, t)$	$\frac{e^{rt}-1}{e^{r}-1}$
Capital Recovery	$(A/P, r, t)$	$\frac{e^{rt}(e^{r}-1)}{e^{rt}-1}$
Sinking Fund	$(A/F, r, t)$	$\frac{e^{r}-1}{e^{rt}-1}$

Excel



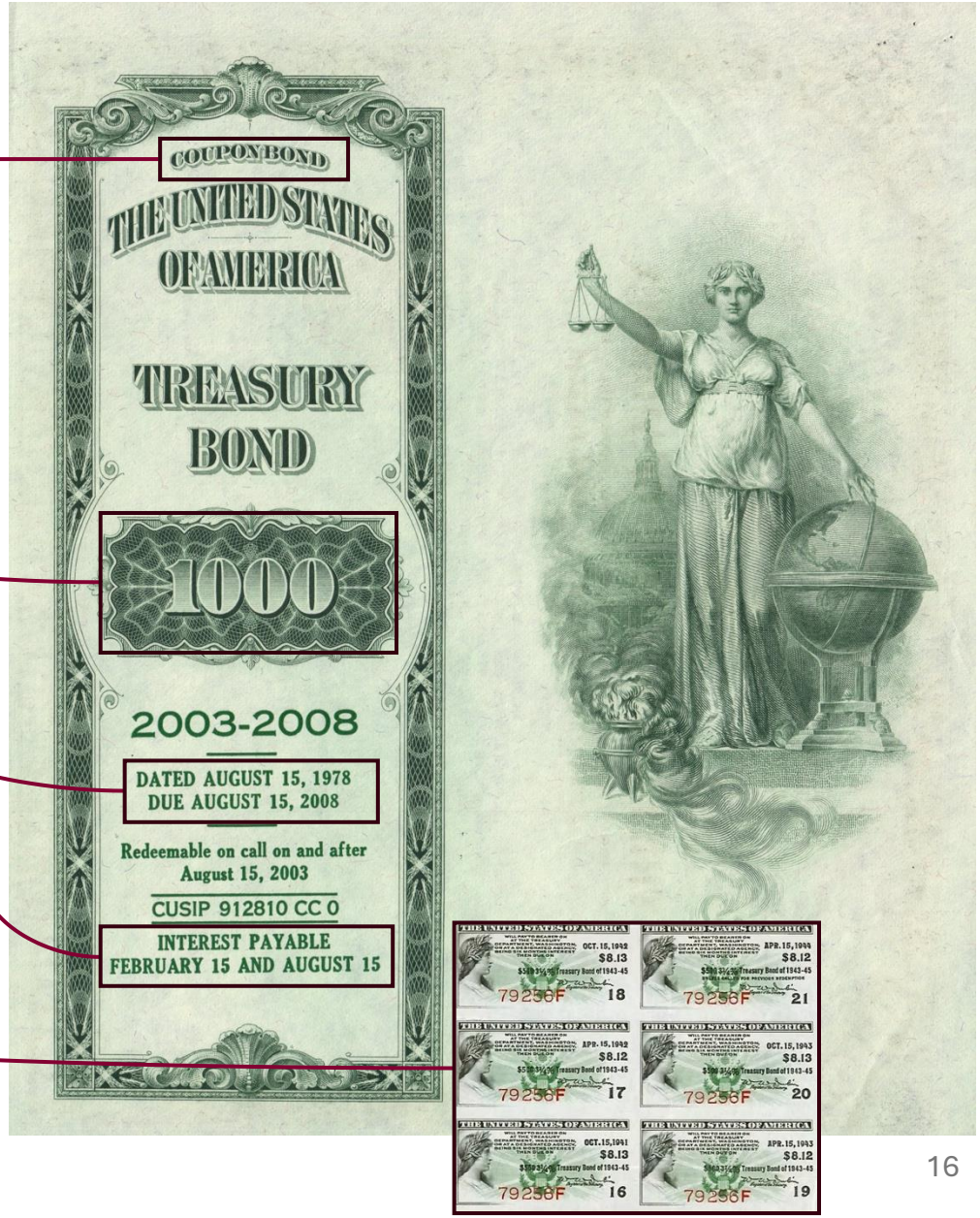
Bonds

A very important Real-world application



Bond

- Coupon Bond
- 1000
- Dated August 15, 1978
  Due August 15, 2008
- Interest Payable February 15
  and August 15
- Coupon



Bond – Important Characteristics

- 1000

Par of Face value

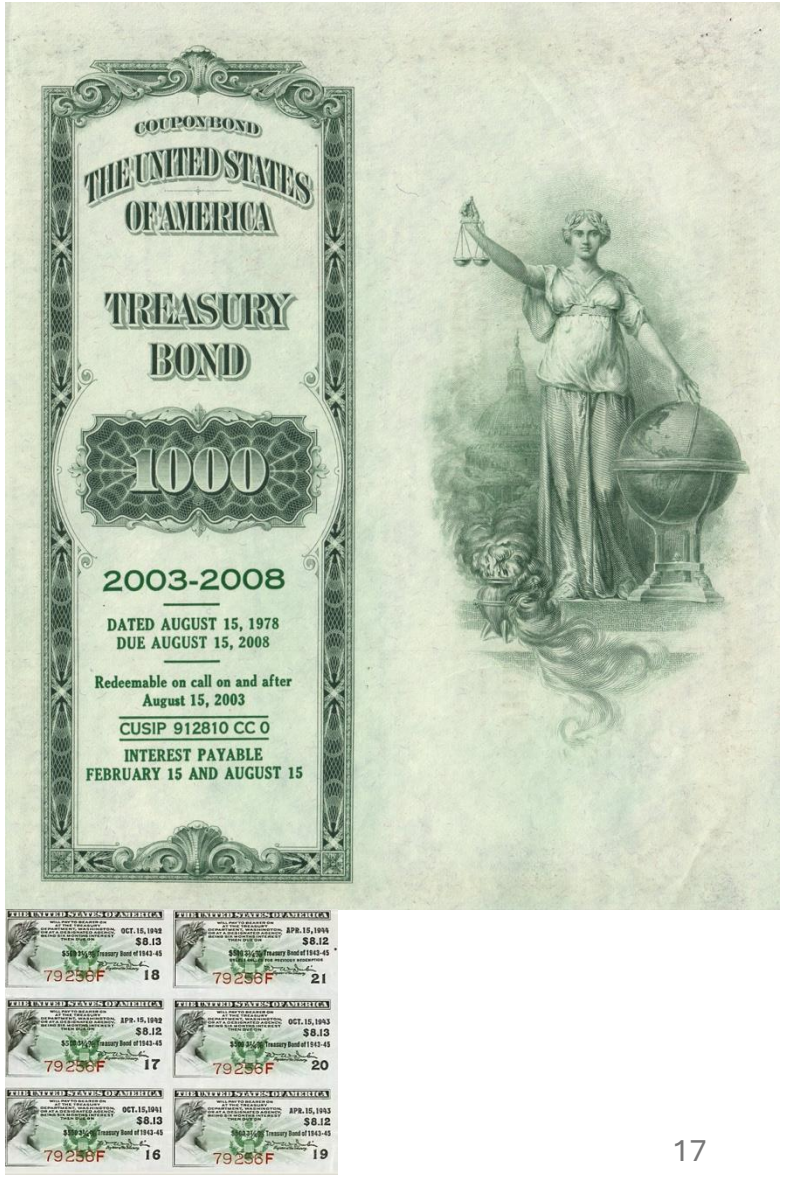
- Dated August 15, 1978
  Due August 15, 2008

Maturity Date

- Coupon

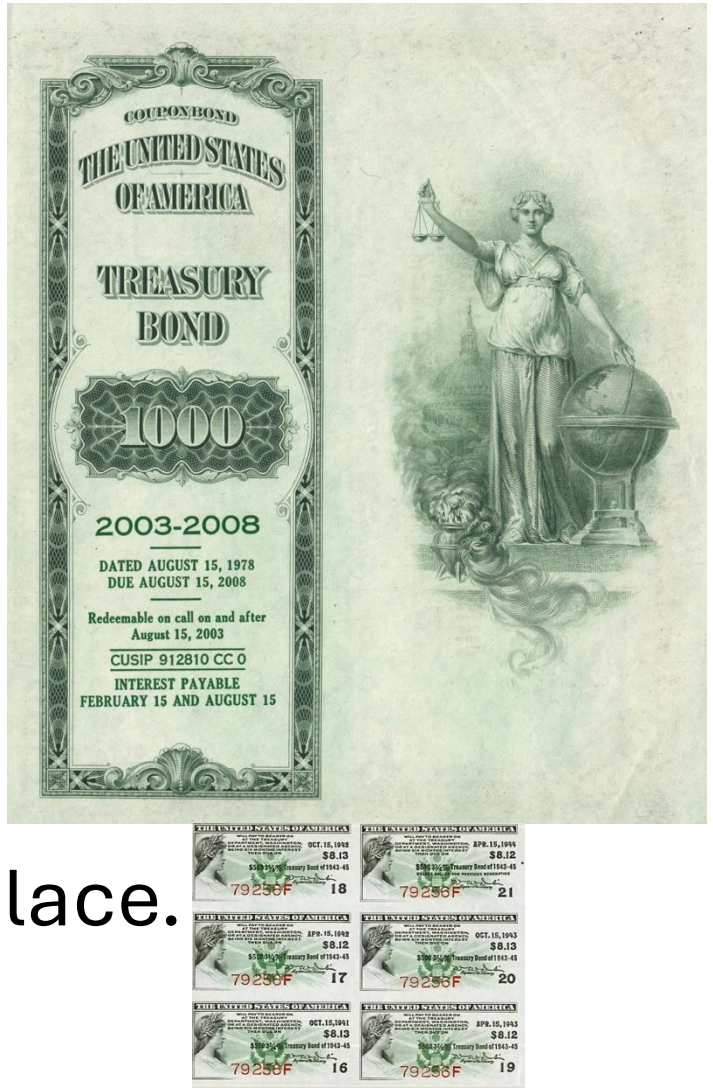
Coupon Amount

Coupon Rate



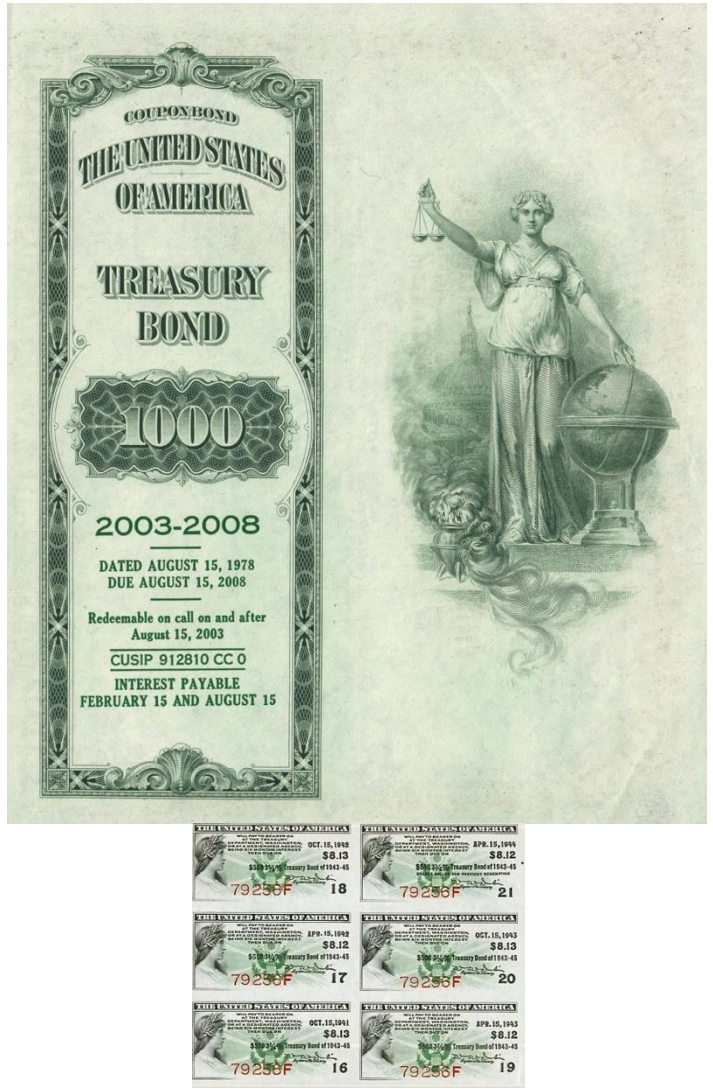
Bond Summary

- **Long-term** note issued by a corporation or a government entity (the borrower) to finance major projects.
- The major source that governments and companies use to obtain **debt financing**
- The borrower receives money **now** in return for a **promise** to pay back to the buyer of the bond
- A bond can be traded freely in the marketplace.
- Its price varies continuously.



Bond - Example

- Assuming the coupon rate is 10%, draw the cashflow of this bond from the point of view of an investor.

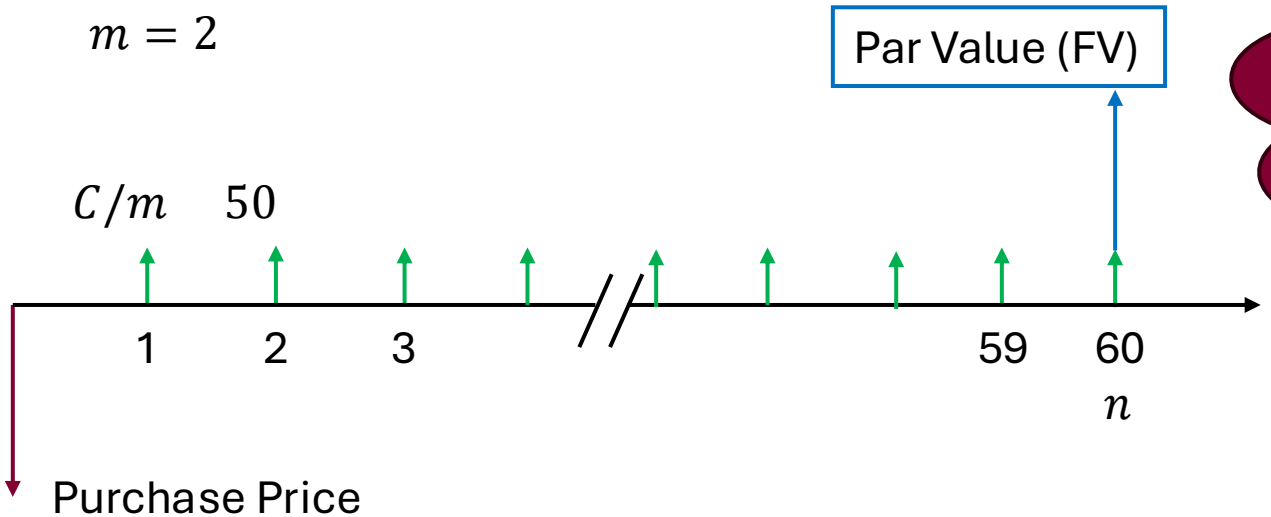


Bond - Example

- Assuming the coupon rate is 10%, draw the cashflow of this bond from the point of view of an investor.

$C = 0.1 \times 1000 = 100$

$m = 2$

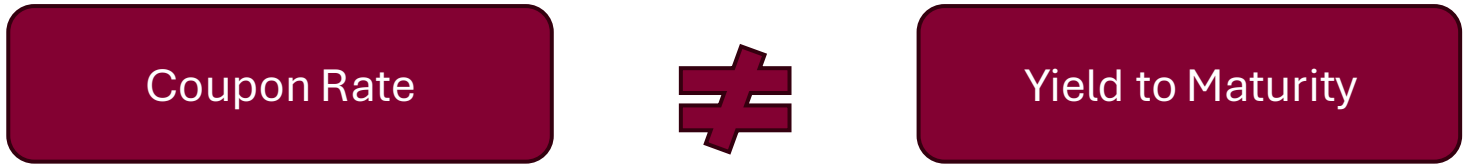


External Market



Bond – Yield to Maturity

- **Yield to Maturity** (YTM) is an **estimate** of the **total rate of return**
  - Quoted on **annual** basis



Internal

Depends on the broader market

- Together they define a potential price for the bond

Bond - Price

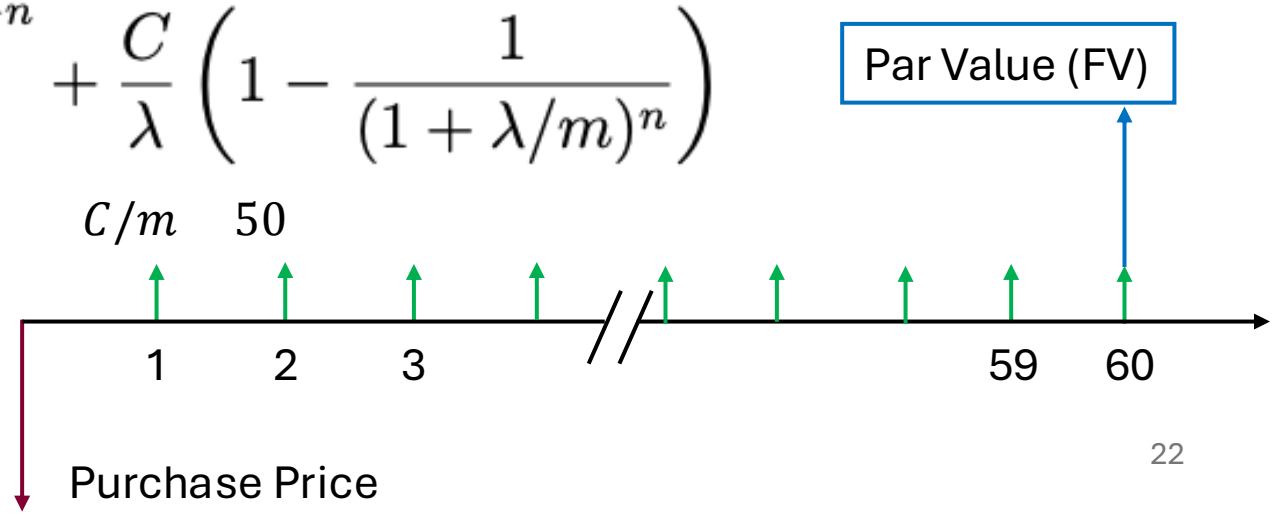
$$PV = F(P/F, \lambda, m, n) + \frac{C}{m}(P/A, \lambda, m, n)$$

OR

$$PV = F(P/F, \lambda/\mathrm{m}, n) + \frac{C}{m}(P/A, \lambda/m, n)$$

OR

$$P = F\left(1 + \frac{\lambda}{m}\right)^{-n} + \frac{C}{\lambda}\left(1 - \frac{1}{(1+\lambda/m)^n}\right)$$



Bond – Yield to Maturity

Year 1

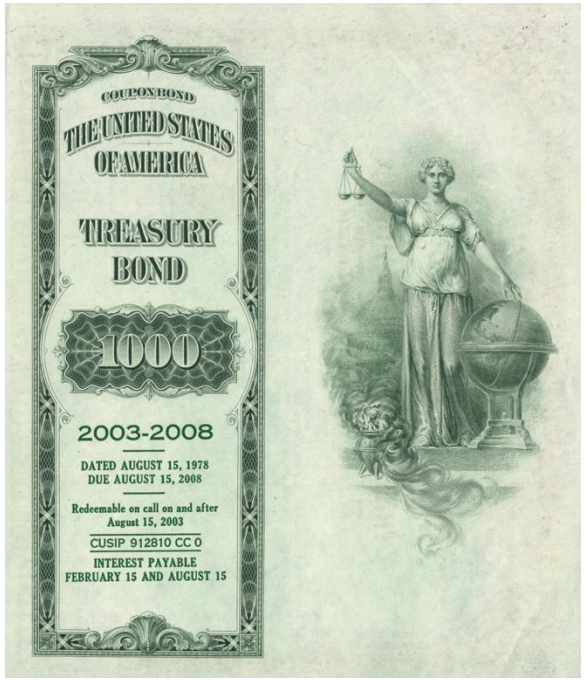








$50 per year
over 10 years



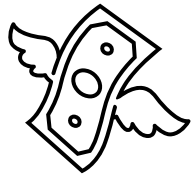
$$PV = F(P/F, \lambda/\mathrm{m}, n) + \frac{C}{m}(P/A, \lambda/m, n)$$
$$=1000(P/F, 0.05{,}10) + 50(P/A, 0.05{,}10)$$
$$=613.91 + 386.09 = 1000$$

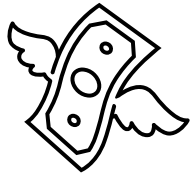
Bond – Yield to Maturity

Year 2

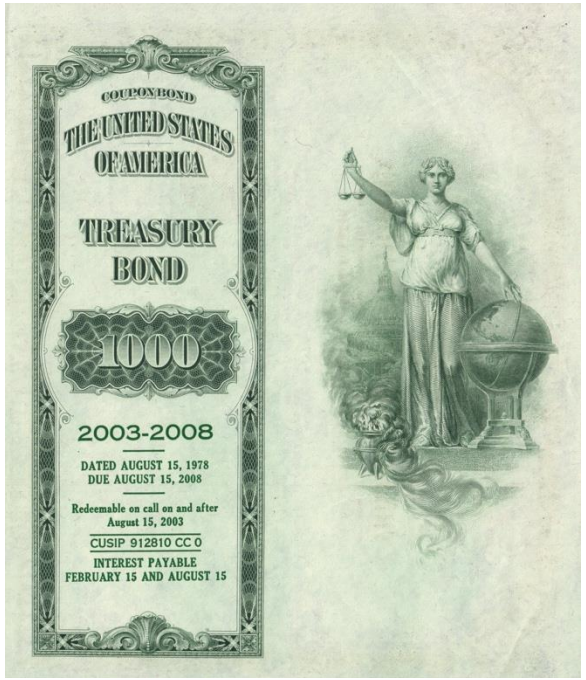








$100 per year
over 10 years

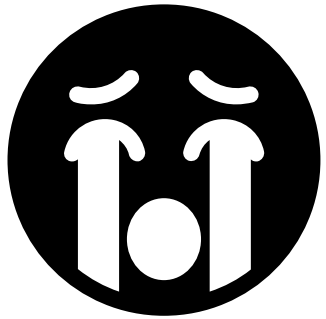


If our investor wants to sell his bond, can they sell it at 1000?

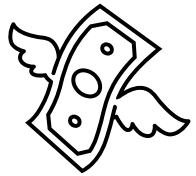
Bonds with new rate

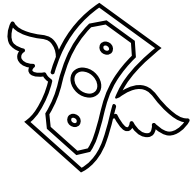
Bond – Yield to Maturity

Year 2

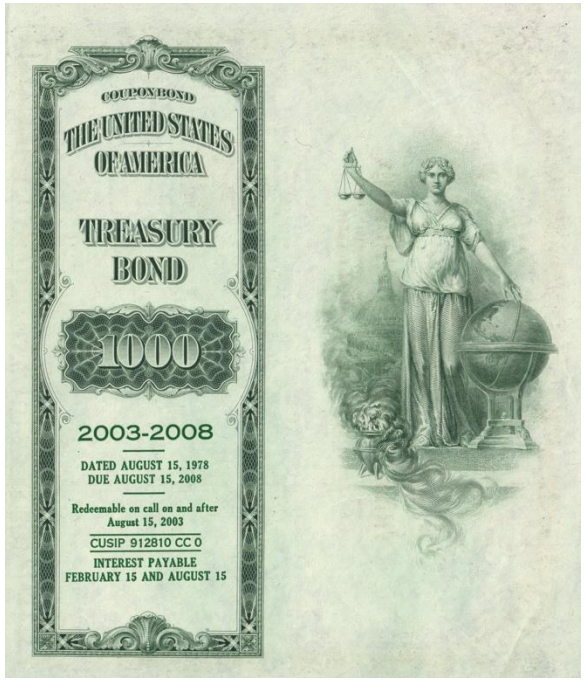








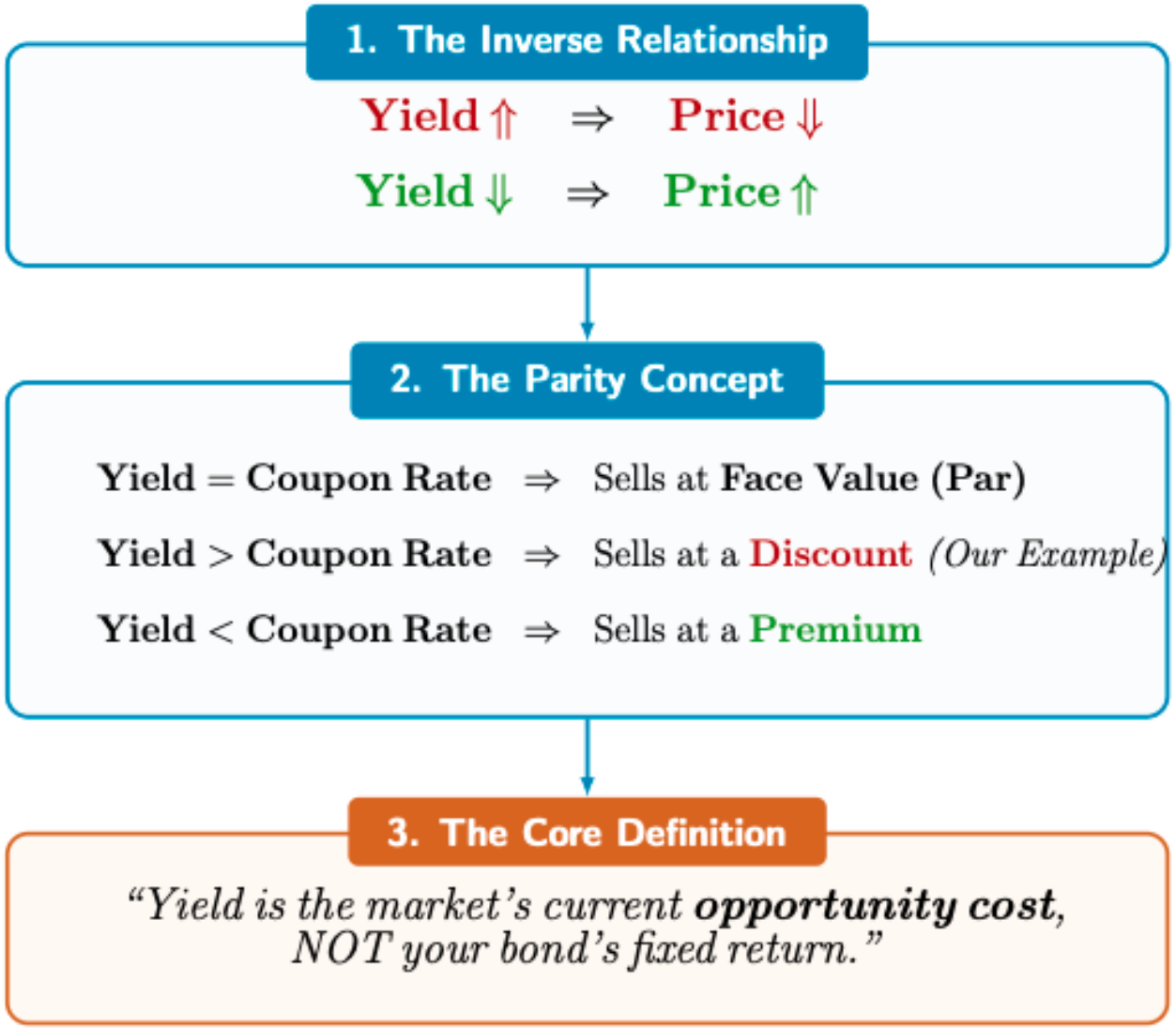
$100 per year
over 10 years



$$PV = F(P/F, \lambda/\mathrm{m}, n) + \frac{C}{m}(P/A, \lambda/m, n)$$
$$=1000(P/F, 0.1{,}9) + 50(P/A, 0.1{,}9)$$
$$=424.1 + 287.95 = 712.05$$

Bonds with new rate

Bond Price - Summary

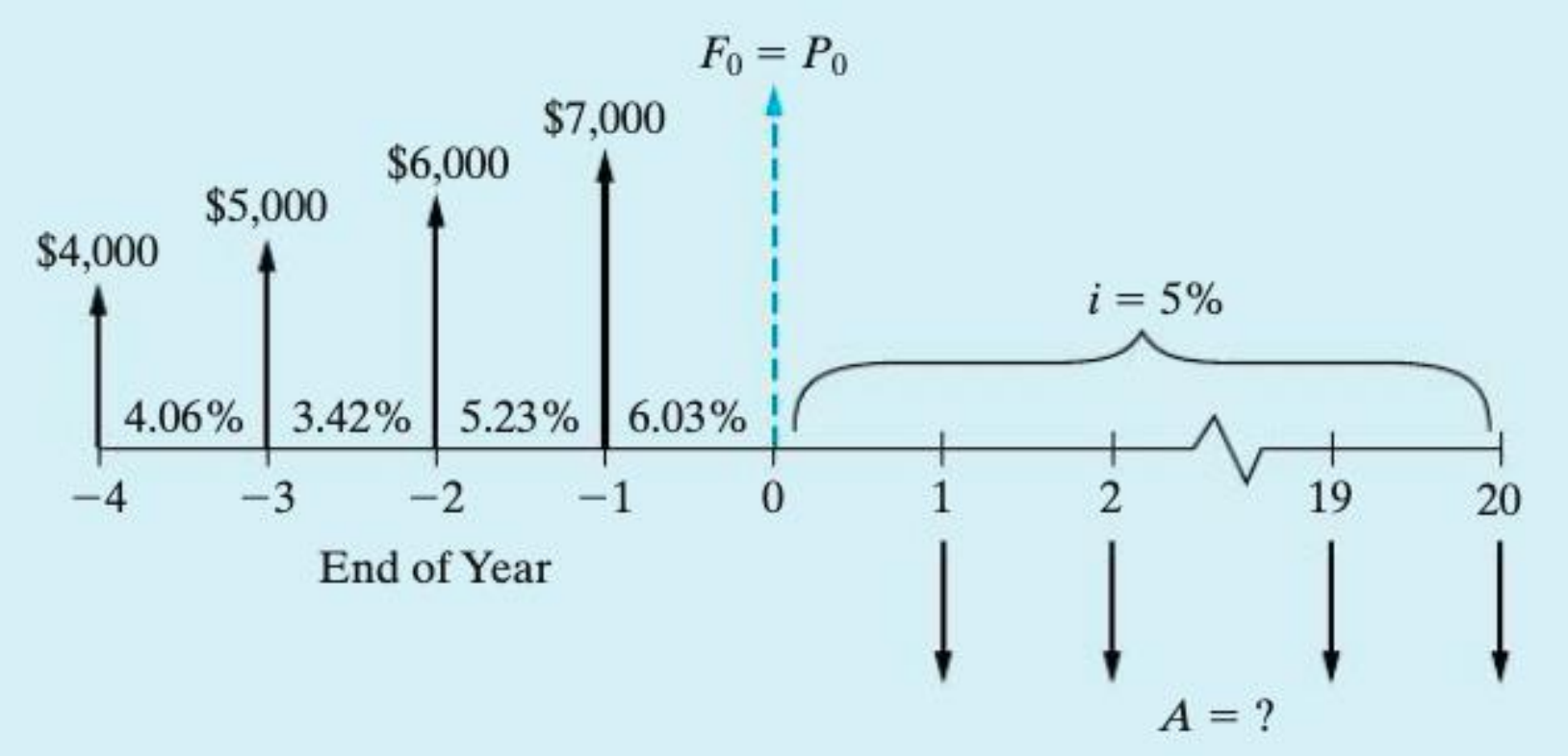


Back to interest rates

Time varying interest rate

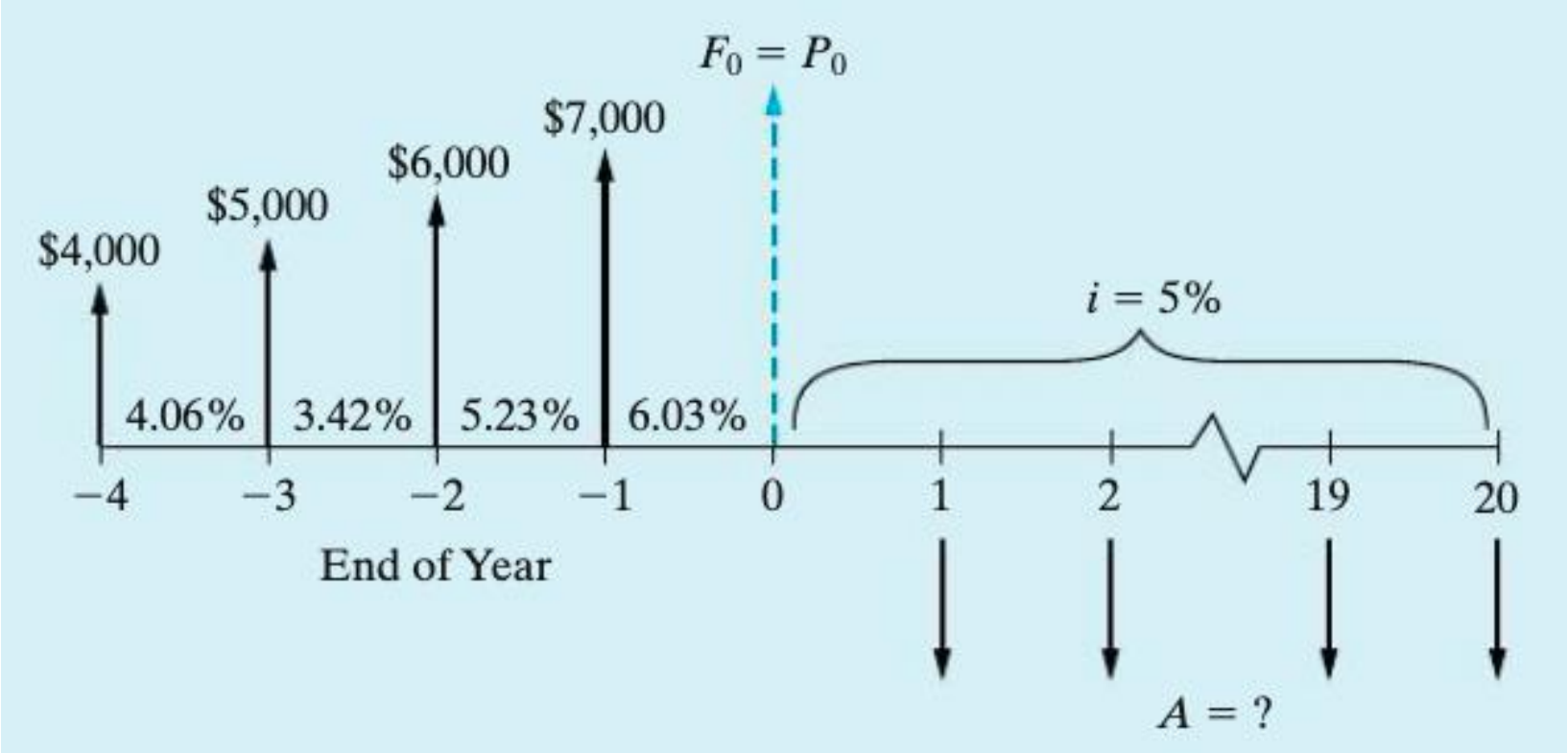


Time Varying Interest Rates - Loan



Stafford student loan: Interest rates are fixed for the life of the loan but are adjusted every July 1 for new borrowers.

Time Varying Interest Rates - Loan



$$F = P(1 + i_1)(1 + i_2) \dots (1 + i_n) = P \prod_{k=1}^{n} (1 + i_k)$$

Time Varying Interest Rates - Investment

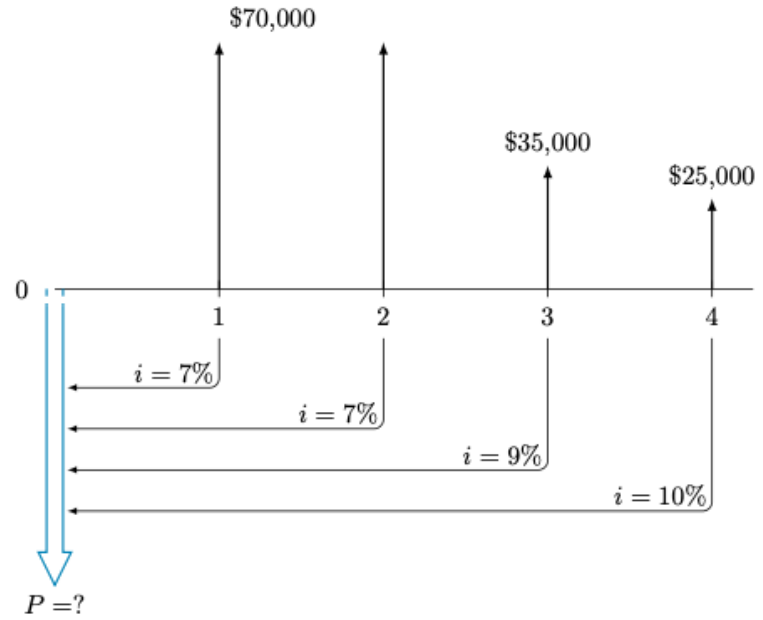
- As a business owner, you would expect to make higher returns with time

Year	1	2	3	4
Net Profit	$70,000	$70,000	$35,000	$25,000
Annual Rate	7%	7%	9%	10%

Time Varying Interest Rates - Investment

- As a business owner, you would expect to make higher returns with time

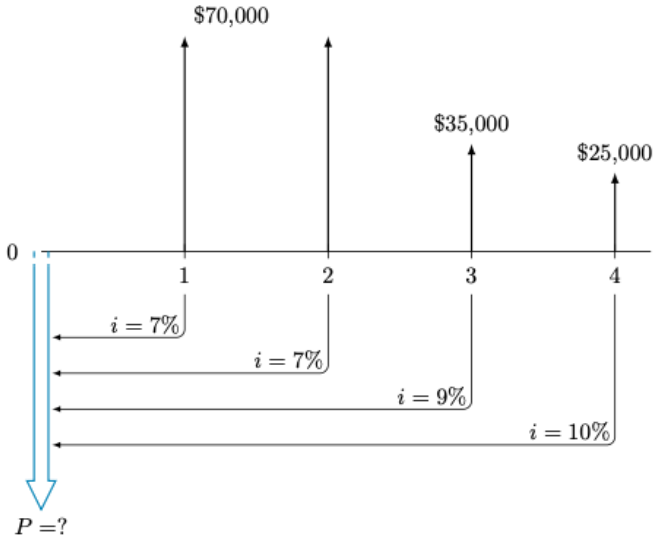
Year	1	2	3	4
Net Profit	$70,000	$70,000	$35,000	$25,000
Annual Rate	7%	7%	9%	10%



Time Varying Interest Rates - Investment

- As a business owner, you would expect to make higher returns with time

Year	Cash Flow	Interest Rate	Denominator (Product of 1+i)	Present Worth (Step)
1	$70,000	7%	1.07	$65,420.56
2	$70,000	7%	1.1449	$61,140.71
3	$35,000	9%	1.2479	$28,046.20
4	$25,000	10%	1.3727	$18,211.82



Time Varying Interest Rates – Annual Equivalence

- https://app.wooclap.com/events/UYEIHQS/questions/6a5e1f629f69834a14c56387

